Enclosure 2 to
NRC-18-0042

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2018-004
Inoperability of Reactor Water Cleanup System Isolation Differential Flow-High Function

NRC FORM 366 (04-2018)

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Fermi 2	05000 341	1 OF 4

4. Title

Inoperability of Reactor Water Cleanup System Isolation Differential Flow-High Function

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name N/A	Docket Number 05000
05	31	2018	2018 -	004 -	00	07	25	2018	Facility Name N/A	Docket Number 05000

9. Operating Mode: 1

10. Power Level: 100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☑ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☑ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Fermi 2 / Scott A. Maglio – Manager, Nuclear Licensing	(734) 586-5076

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
B	CE	CAP	X999	Yes	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date)	☑ No				

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On May 31, 2018 at 1420 EDT, the Reactor Water Cleanup (RWCU) System Isolation Differential Flow - High function was declared inoperable as a result of the RWCU Differential Flow Indicator reading downscale. This condition would have prevented the primary containment isolation valves for the RWCU system from automatically isolating on a high differential flow signal. The cause of the downscale reading was due to a capacitor failure within the square root converter. Corrective actions were taken to replace the capacitor within the square root converter and return the RWCU System Isolation Differential Flow - High function to operable on June 1, 2018 at 0947 EDT. Additional corrective actions included verification of the shelf life of the current inventory of capacitors and a 10-year preventative maintenance activity to replace the capacitor due to the 10-year service life. The safety related function of the RWCU System is to provide primary containment isolation upon receipt of a primary containment isolation signal, or upon indication of a high-energy line break (HELB) in the RWCU System piping in response to the following signals: low reactor water level 2; high RWCU equipment temperatures; high RWCU pump room differential temperature; or high RWCU System differential flow. Due to the availability of other automatic RWCU isolation functions as well as the ability to manually isolate RWCU, the safety significance of this event is very low. There were no radiological releases associated with this event.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Fermi 2	05000- 341	2018	- 004	- 00

NARRATIVE

INITIAL PLANT CONDITIONS
Mode – 1
Reactor Power – 100 percent

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT
On May 31, 2018 at 1420 EDT, the Reactor Water Cleanup (RWCU) [[CE]] System Isolation Differential Flow - High function was declared inoperable as a result of the RWCU Differential Flow Indicator [[FI]] reading downscale; which would have prevented the primary containment isolation valves [[ISV]] for the RWCU system from automatically isolating on a high differential flow condition. At 1519, RWCU was shutdown and the affected penetration flow paths were isolated in accordance with station procedures as required by Fermi 2 Technical Specification 3.3.6.1. All other RWCU primary containment isolation instrumentation functions remained operable and the associated RWCU system primary containment isolation valves were capable of being remotely closed by the control room operators throughout the event.

The high differential flow signal is initiated from transmitters that are connected to the RWCU pump outlet and RWCU system discharge to condenser and feedwater. The outputs of the transmitters are compared (in a common summer) and the resulting output is sent to two high flow trip systems. If the difference between the inlet and outlet flow is too large, each trip unit generates an isolation signal. Inoperability of the non-redundant circuitry causes the channels in both trip systems to be inoperable. The remainder of the circuit is redundant and can be considered on a per trip system basis. The square root converter [[CNV]] is within the non-redundant portion of this string.

The RWCU system is non-safety; however, the isolation valves provide a safety function to provide primary containment isolation upon receipt of a primary containment isolation signal, or upon indication of a high-energy line break (HELB) in the RWCU system piping in response to the following signals; low reactor water level 2; high RWCU equipment temperatures; high RWCU pump room differential temperature; or high RWCU system differential flow. The differential flow isolation signal is designed to detect the leaks below the capacity of the reactor coolant makeup and within the cold leg portion (approx. 120F) of the system, which would not change ambient air conditions.

A similar event had occurred previously on May 27, 2018 as described in LER 2018-003. Actions had been taken to replace a capacitor [[CAP]] within the square root converter. Troubleshooting during this event identified that the replacement capacitor had failed. Following another replacement of the capacitor within the square root converter, the RWCU System Isolation Differential Flow - High function was declared operable on June 1, 2018 at 0947 EDT. Therefore, the non-redundant RWCU System Isolation Differential Flow - High function was inoperable for approximately 19.5 hours.

Event Notification 53435 was made at 1620 on May 31, 2018 pursuant to 10CFR 50.72 (b)(3)(v)(C) and 10CFR 50.72 (b)(3)(v)(D). This report is being made pursuant to 10CFR50.73(a)(2)(v)(C) as a condition that could have prevented the fulfillment of a safety function needed to control the release of radioactive material and 10CFR50.73(a)(2)(v)(D) as a condition that could have prevented the fulfillment of a safety function needed to mitigate the consequences of an accident. The Updated Final Safety Analysis Report (UFSAR), the TS Bases, and NRC reporting requirements were reviewed as part of an evaluation regarding event reportability. Based on this review, it was determined that the RWCU Differential Flow leak detection function is credited to mitigate a break in the RWCU system piping accident as described in UFSAR Chapter 6.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Fermi 2	05000- 341	2018	- 004	- 00

NARRATIVE

Technical Specification 3.3.6.1, Primary Containment Isolation requires the RWCU Differential Flow-High to be operable in Modes 1, 2, and 3. The high differential flow signal is provided to detect a break in the RWCU System. This will detect leaks in the RWCU System when area or differential temperature would not provide detection (i.e., a cold leg break). This Function is not assumed in any UFSAR transient or accident analysis, since bounding analyses are performed for large breaks such as Main Steam Line Breaks. However, UFSAR Section 6.2.4.2.6, Leak Detection, states "For systems penetrating the primary containment, major leaks in the pipe are located by increased temperature, radiation, sump level, changes in pressure, differential pressure, process line flow, etc. These indications are monitored in the control room to alert the operator when remote manual valves should be closed. In addition, certain indications of leakage will cause automatic valves to close in response to a system accident."

A review was performed for the time the RWCU differential flow high function was previously restored (May 28 at 0046) until this failure (May 31 at 1420). Based on the review it was determined that this function was operable until failure on May 31, 2018 at 1420.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

Within the RWCU, system leakage is generally detected by high temperature sensing instrumentation which senses a leak before a line break by monitoring the temperature rise in the equipment compartment for a system. This is based upon having the leak within the hot portion of the system. However, temperature sensing does not detect leakage from the cold-water part of the RWCU System. In the Differential Flow Leak Detection System, leakage is detected by means of a flow comparison between RWCU system inlet and outlet. If the inlet flow exceeds outlet flow by approximately 55 gpm an alarm is actuated and the RWCU system is isolated automatically.

The RWCU isolation function (low reactor water level) and all other RWCU isolation functions remained operable throughout this event. In addition, the associated RWCU system primary containment isolation valves were capable of being remotely closed by the control room operators throughout this event. Based on this discussion, the safety significance of this event is very low. There were no radiological releases associated with this event.

CAUSE OF THE EVENT

The cause of this event was due to a lost signal from a flow input due to a failed capacitor within the square root converter. This failure occurrence has been attributed to the replacement C5 capacitor that was taken from the spare parts inventory, and was beyond the usable shelf life causing failure after the square root converter was placed in-service.

As described in LER 2018-003, the capacitor in the square root converter had been replaced on May 28, 2018 with one that was beyond its usable shelf life and functioned until May 31, 2018 when it was observed that flow indication had been lost.

Upon further review of the May 31, 2018 event, it was identified that the square root converter had been replaced following the December 11, 2017 performance of a surveillance procedure where the RWCU Inlet Flow square root converter was found to be out of tolerance and could not be recalibrated.

Cause: Capacitor
System: G33 – Reactor Water Cleanup
Component: G33K602
Manufacturer: Generic Component

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Fermi 2	05000- 341	2018	004	00

NARRATIVE

CORRECTIVE ACTIONS

Immediate corrective actions were taken to replace the capacitor within the square root converter and return the RWCU System Isolation Differential Flow - High function to operable on June 1, 2018 at 0947 EDT. Before replacement, the replacement capacitor was verified to be within its usable shelf life to eliminate the potential for recurrence.

In addition, the plant verified the shelf life of the current inventory of capacitors and a 10-year preventative maintenance activity to replace the capacitor due to the 10-year service life was developed following the May 31, 2018 event.

PREVIOUS OCCURRENCES

LER 2018-003 documented a similar failure on May 27, 2018 at 0630 EDT. As described above, the corrective actions taken in response to LER 2018-003 (i.e. to replace the capacitor) did not prevent this LER since the replacement capacitor was beyond its useful life.